# APS110H1f2026 - October 1, 2026

October 1, 2026 8:01 AM

- Properties of Matter(s) - today again (and also for this week as well again) - also - "I Need More Structure in My Life" plus "Even More Structure"
  - Calculation +
  - BCC - Body Centred Cubic Crystal ---> FCC Face Centred Cubic
  - Nano structure - bonding
  - Another nano structure - maybe

BCC --> FCC

October 1, 2026 8:08 AM

BCC- Body Centred Cubic Crystal.

FCC- Face Centred Cubic

metal

$$APF = \frac{n \frac{4}{3} \pi R^3}{a^3}$$

different for BCC

$$APF_{BCC} = \frac{2 \frac{4}{3} \pi R^3}{\left(\frac{4}{\sqrt{3}} R\right)^3}$$

R= pm in the periodic table

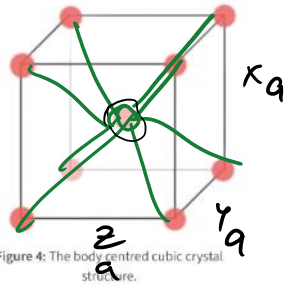



Figure 4: The body centred cubic crystal structure.

Figure 4

$n_{BCC} = 2$

BCC - n = 2

BCC - a $= \frac{4}{\sqrt{3}} R$

$a_{BCC} = \frac{4}{\sqrt{3}} R$

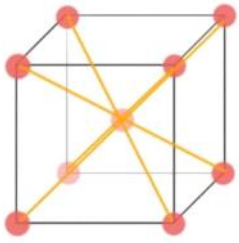


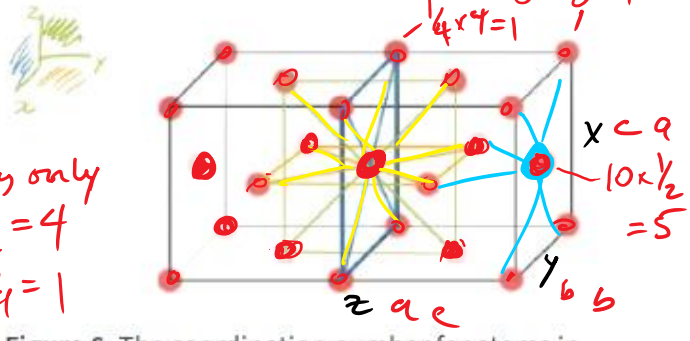
Figure 5: The body centred cubic crystal structure, showing the direction of contact across the cube diagonals and the coordination number of eight.

Figure 5

metals.

8 x 1/8 = 1

1/4 x 4 = 1




→ if bonding only

8 x 1/2 = 4

4 x 1/4 = 1

4 + 1 = 5

→ + others / not bonded.

8 x 1/8 = 1

2 x 1/2 = 1

**Figure 6:** The coordination number for atoms in FCC. The right side face-centred atom makes contact with four atoms in the right side plane, with four atoms in the face-centred positions in the original unit cell, and with four atoms in teh face-centred positions in the unit cell to the right.

coordination number FCC

$CN_{FCC}$

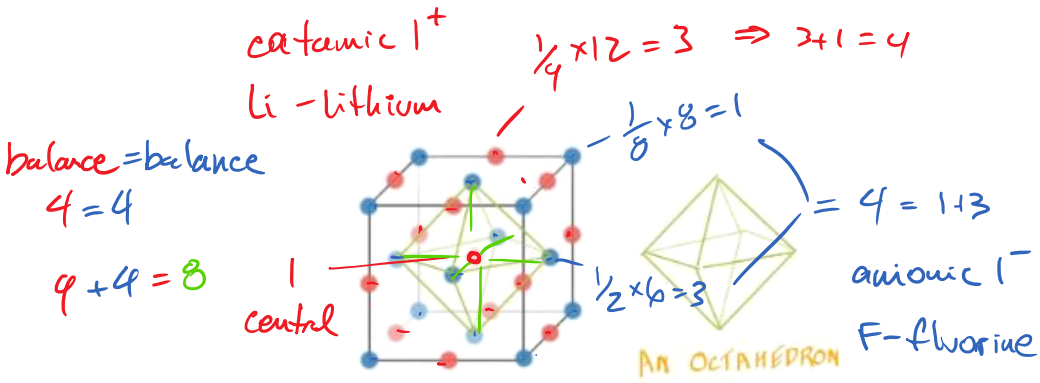



**Figure 7:** The octahedral interstitial site. Note that all cations, not just the central one, occupy octahedral interstitial sites within rock salt.

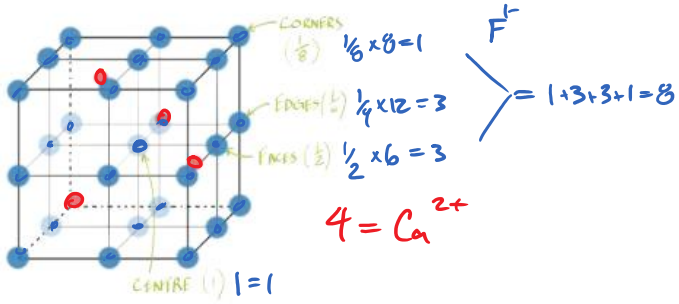



**Figure 8.** There are 8 anions within the calcium fluorite ceramic crystal structure. The fraction of an atom for each position type is shown here.

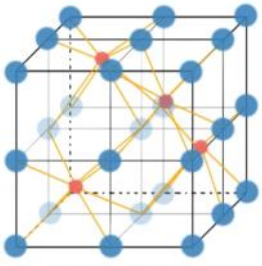


**Figure 6.** The calcium fluorite ceramic crystal structure. Anions represented as blue dots are in a simple cubic lattice with half of the simple cubic interstitial sites occupied by cations. Cations touch their 8 nearest neighbour anions along the cube diagonals, as illustrated by the orange lines.

better manufacturing "maybe"

"sin" = the size of interstital sites

R = radius

$$\frac{R_c}{R_A} = \frac{1 - \sin 45}{\sin 45} = \frac{1 - \frac{1}{1.414}}{\frac{1}{1.414}}$$

$$\Downarrow$$

$$\sin 45 = \frac{2R_A}{2R_A + 2R_c}$$

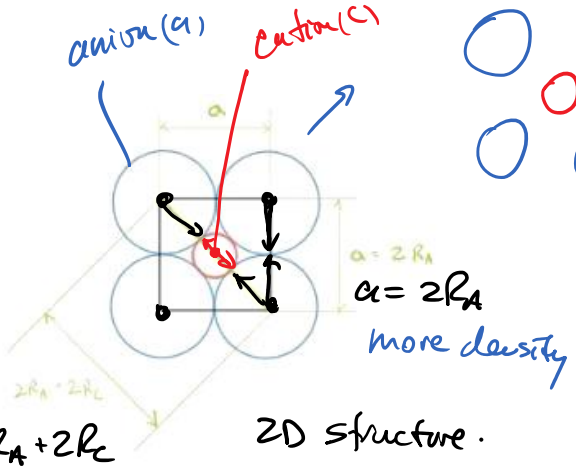



**Figure 9:** A 2D slice through the octahedral interstitial site at the geometrically ideal radius ratio when the anions are touching one another and the cation is just big enough to fit into the site without pushing the anions apart.